

Fundamentals of Machine Learning: A Focus on Regression and Regularization

1. Introduction to Machine Learning (ML)

Machine Learning (ML) is a subfield of artificial intelligence (AI) that focuses on building systems capable of learning from data, identifying patterns, and making decisions with minimal human intervention. Unlike traditional programming, where explicit rules are coded to solve a problem, ML algorithms learn these rules directly from the data.

The core premise is to build a mathematical model based on sample data, known as "training data," to make predictions or decisions without being explicitly programmed to do so. As the models are exposed to new data, they are able to independently adapt, uncovering previous or newly emerging patterns.

Machine Learning is broadly categorized into three main types:

1. **Supervised Learning:** The algorithm is trained on a labeled dataset, meaning the input data is paired with the correct output. The model learns the mapping function from inputs to outputs.
2. **Unsupervised Learning:** The algorithm is given unlabeled data and must find the underlying structure or patterns within it on its own.
3. **Reinforcement Learning:** The algorithm learns by interacting with an environment, receiving rewards or penalties for its actions, aiming to maximize the cumulative reward.

This report will focus on two foundational supervised learning algorithms—Linear Regression and Logistic Regression—and a crucial technique for improving model performance called Regularization.

2. Linear Regression

Linear Regression is perhaps one of the most well-known and well-understood algorithms in statistics and machine learning. It is a supervised learning algorithm used for predicting a continuous numerical value.

Concept

The goal of linear regression is to model the relationship between a dependent variable (the target we want to predict) and one or more independent variables (the features). It does this by fitting a linear equation to the observed data.

- **Simple Linear Regression:** Involves one independent variable. The relationship is modeled as a straight line where the dependent variable is calculated using the independent variable, a y-intercept, a slope (coefficient), and an expected error term.
- **Multiple Linear Regression:** Involves two or more independent variables. The relationship is modeled as a hyperplane where multiple independent variables, each with their own assigned coefficient, combine to predict the target outcome.

The algorithm "learns" by finding the optimal values for the coefficients. The most common method to achieve this is **Ordinary Least Squares (OLS)**. OLS calculates the difference between the actual data points and the predicted values on the line (these differences are called residuals) and seeks to minimize the sum of the squared residuals.

By minimizing this error, the algorithm finds the line of "best fit" that represents the general trend of the data.

Applications

Linear regression is used when the output variable is continuous. Examples include:

- Predicting house prices based on square footage, location, and number of bedrooms.
- Forecasting sales revenue based on advertising spend.
- Estimating a person's life expectancy based on health indicators.

3. Logistic Regression

Despite its name containing the word "regression," Logistic Regression is actually a powerful algorithm used for **classification** tasks, not continuous prediction. It is widely used when the dependent variable is categorical, most commonly binary (two classes).

Concept

Instead of predicting a continuous value like linear regression, logistic regression predicts the probability that an instance belongs to a specific class. Because it predicts a probability, the output must be between 0 and 1.

To achieve this, logistic regression uses the **Logistic Function** (also known as the Sigmoid function). The function takes any real-valued number and maps it into a value between 0 and 1. It is defined such that it maps large positive values close to 1 and large negative values close to 0.

In logistic regression, the model calculates a combined score based on the features and their learned weights. It then passes this score through the sigmoid function to output a final probability.

Decision Boundary

Once the probability is calculated, a threshold is applied to make the final classification. Typically, the threshold is set at 0.5.

- If the predicted probability is greater than or equal to 0.5, the instance is classified into Class 1.
- If the predicted probability is less than 0.5, the instance is classified into Class 0.

Applications

Logistic regression is used for binary classification problems (though it can be extended to multi-class problems). Examples include:

- **Spam Detection:** Classifying an email as "Spam" or "Not Spam".

symptoms.

- **Credit Scoring:** Determining if a customer is likely to default on a loan.

4. Regularization

When training machine learning models, a common problem is **overfitting**. Overfitting occurs when a model learns the training data too well, capturing the noise and random fluctuations rather than the underlying pattern. An overfitted model performs exceptionally well on the training data but poorly on new, unseen data.

Regularization is a set of techniques used to prevent overfitting by discouraging the model from becoming too complex. It achieves this by adding a "penalty" term to the loss function that the model is trying to minimize.

The total loss function becomes the standard error (the difference between the model's predictions and actual data) plus a penalty for model complexity. A hyperparameter (often referred to as lambda) controls the strength of the regularization. A higher value means more regularization, leading to a simpler model.

Types of Regularization in Regression

The two most common types of regularization applied to regression models are Ridge Regression (L2) and Lasso Regression (L1).

4.1. Ridge Regression (L2 Regularization)

Ridge regression adds a penalty term proportional to the **square of the magnitude** of the coefficients.

- **Effect:** It shrinks the coefficients of less important features close to zero, but it rarely forces them exactly to zero. This helps in reducing model complexity and handling multicollinearity (when independent variables are highly correlated).

4.2. Lasso Regression (L1 Regularization)

Lasso (Least Absolute Shrinkage and Selection Operator) regression adds a penalty term proportional to the **absolute value of the magnitude** of the coefficients.

- **Effect:** Unlike Ridge, Lasso has the ability to shrink coefficients completely to zero. This means it acts as a form of **feature selection**, effectively removing less important features from the model entirely. This is particularly useful when dealing with datasets that have a large number of features.

Summary of Regularization

Regularization is crucial for building models that generalize well to new data. By applying L1 or L2 regularization, data scientists can control the complexity of Linear and Logistic regression models, ensuring they remain robust and reliable.

5. Conclusion

learning, handling continuous and categorical outcomes, respectively. While powerful, these models are susceptible to overfitting, especially when dealing with complex datasets. Regularization techniques like Ridge (L2) and Lasso (L1) are essential tools for managing this complexity, improving model generalizability, and ensuring that ML solutions remain effective when deployed in real-world scenarios.